

EXPERIMENT : 22

To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

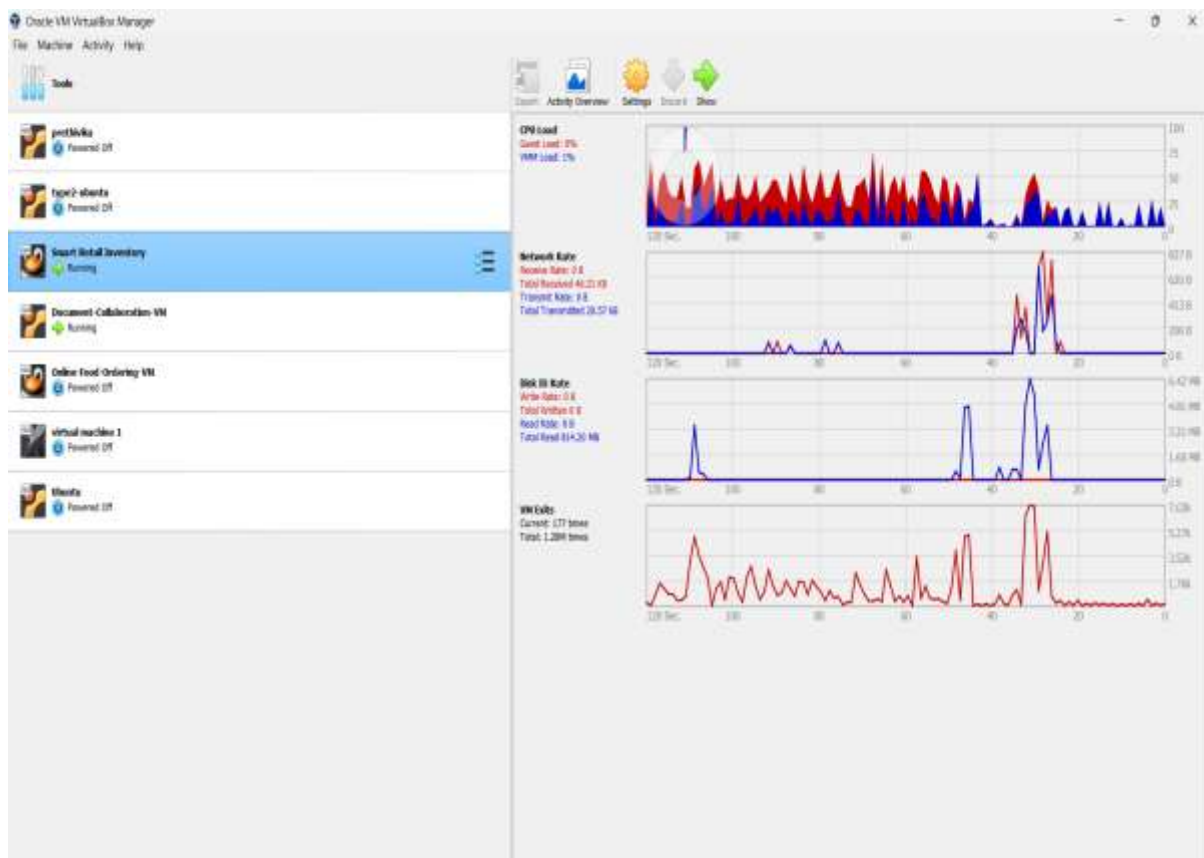
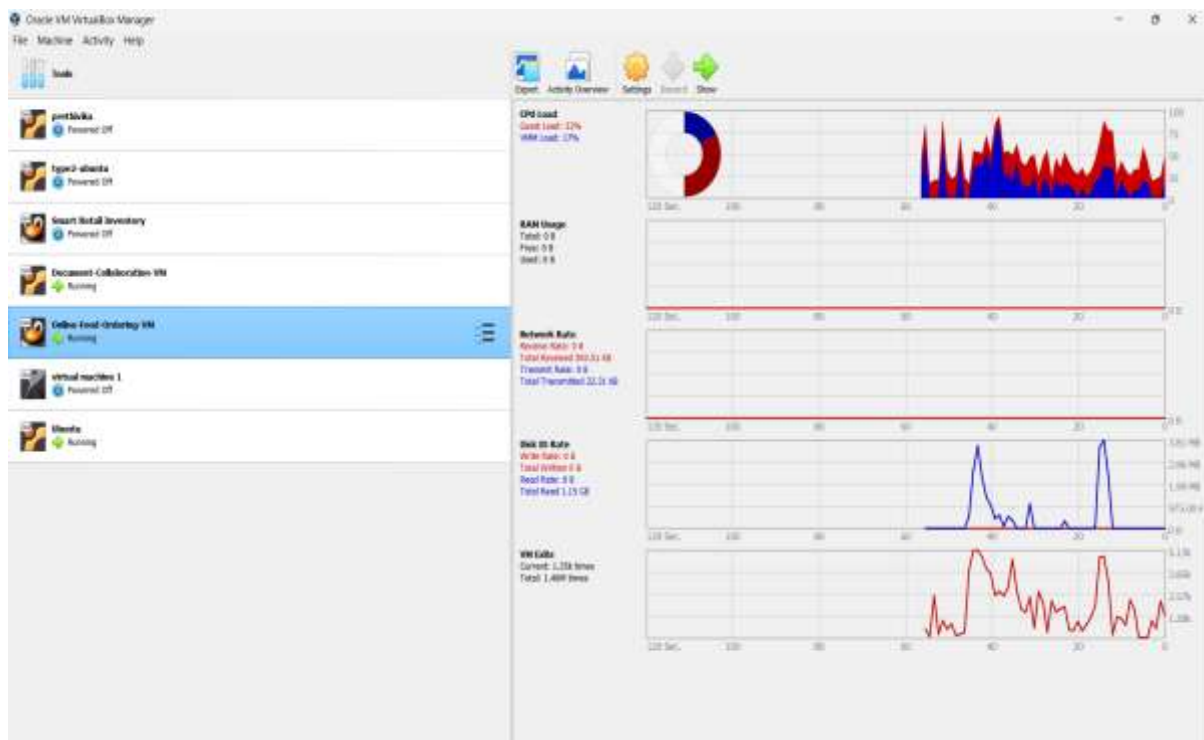
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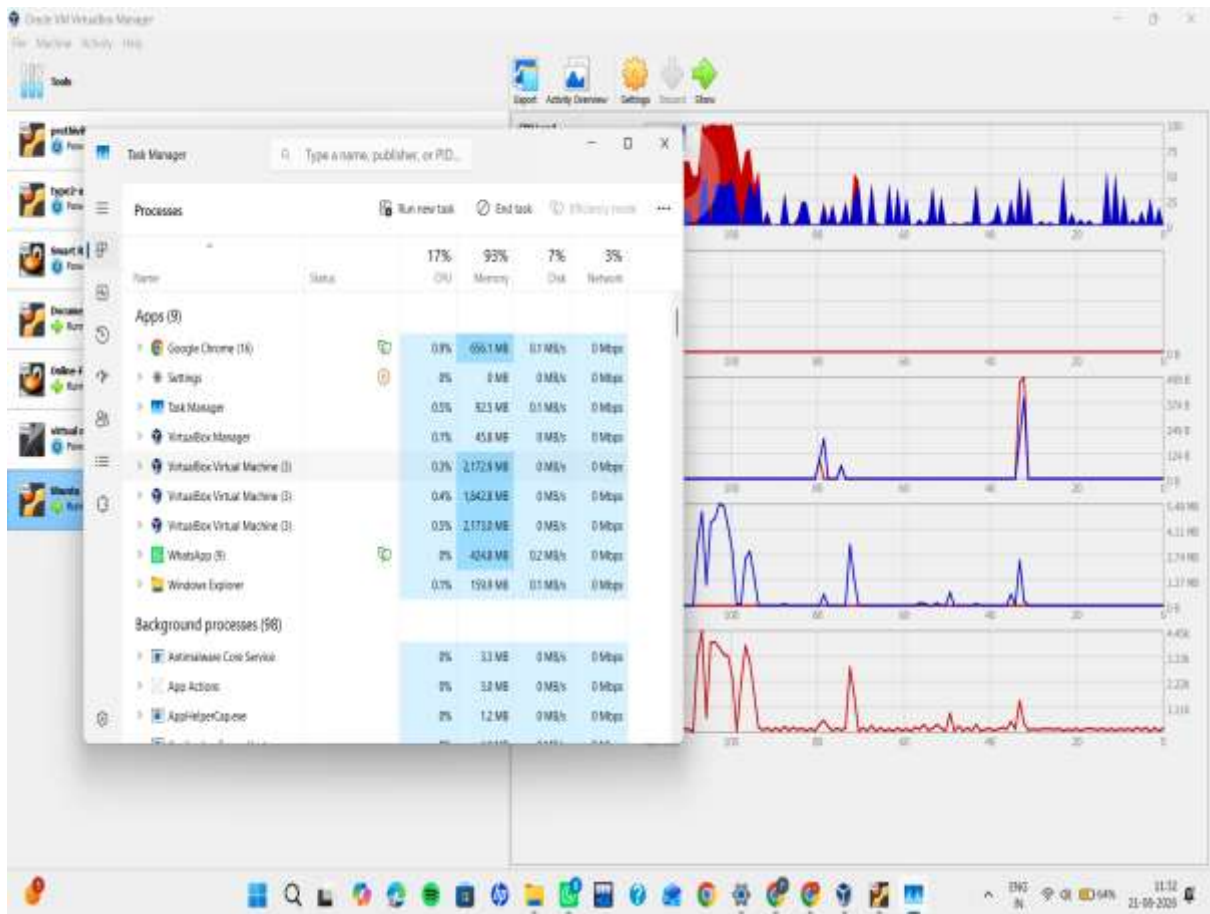
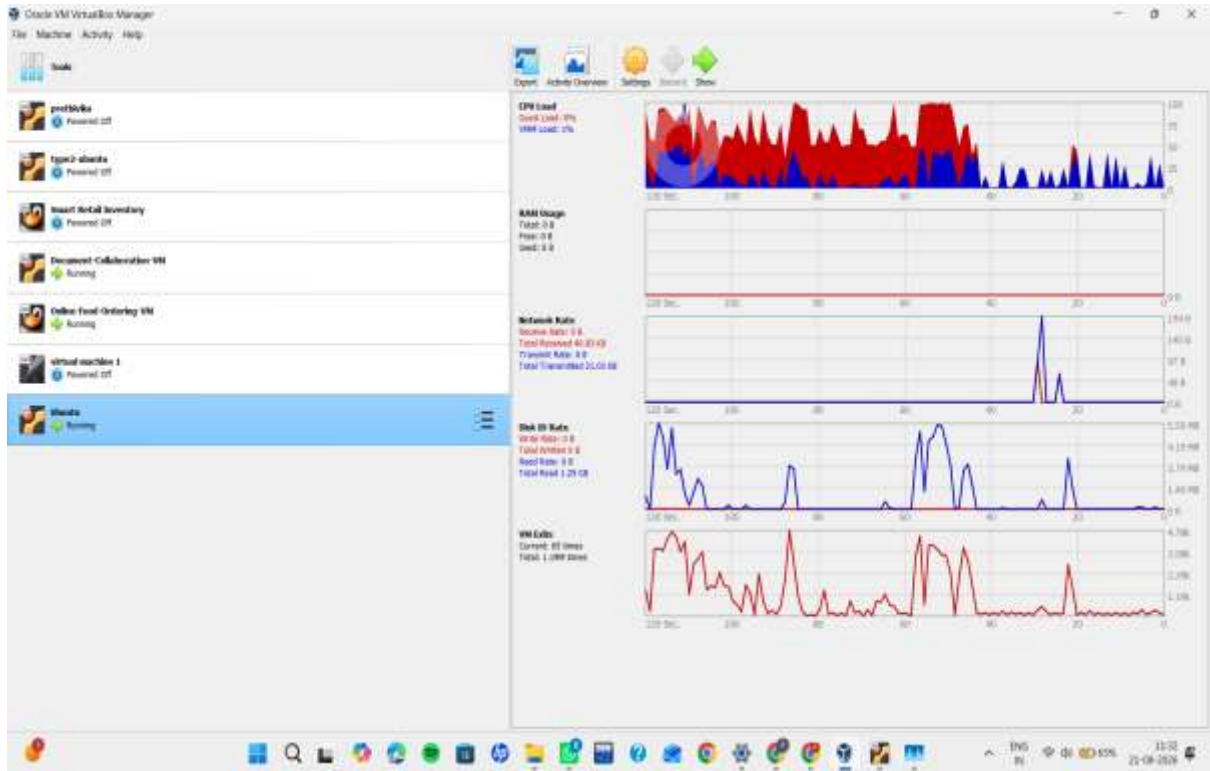
To create and run virtual machines with different hardware configurations using Oracle VirtualBox and determine how many virtual machines can be utilized simultaneously based on the available CPU, memory, and storage resources of the host system.

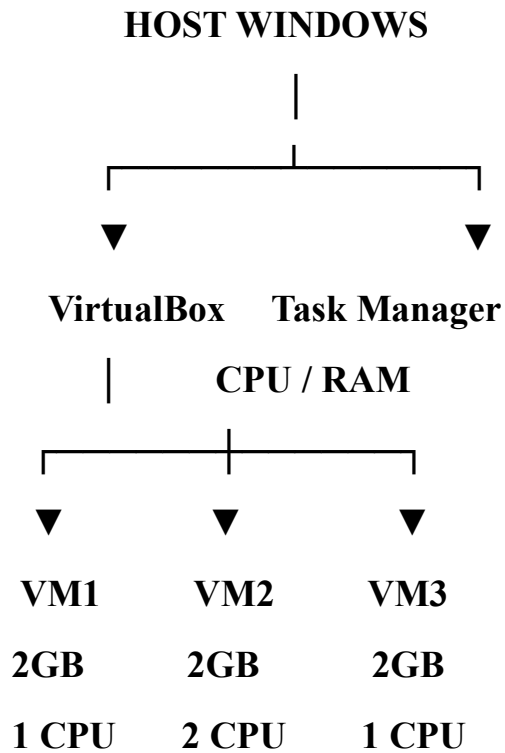
PROCEDURE

1. Open **Oracle VirtualBox** on your Windows system and check the available **RAM, CPU cores, and storage** of the host computer.
2. Create the first virtual machine by selecting **New**, choose Ubuntu/Linux as the guest OS, and allocate a configuration such as **2 GB RAM and 1 CPU**.
3. Create a second virtual machine with a different configuration, for example **2 GB RAM and 2 CPUs**, and install or attach the required Linux ISO.
4. Create a third virtual machine with another configuration, such as **4 GB RAM and 2 CPUs**, if sufficient host resources are available.
5. Start the virtual machines one at a time and then start multiple VMs simultaneously. Open **Task Manager** → **Performance** in Windows to observe the CPU and memory utilization of the host system.
6. Record the number of virtual machines that can run simultaneously without excessive CPU usage, memory exhaustion, or significant performance degradation. For example, if three VMs run smoothly but starting a fourth causes very high memory or CPU utilization, consider three VMs as the practical limit for the selected configurations.
7. Stop the virtual machines and compare their configurations and resource utilization. Determine the maximum number of VMs that can be effectively utilized at the same time based on the available host resources.

IMPLEMENTATION:







RESULT

Virtual machines with different CPU and memory configurations were successfully created and executed using **Oracle VirtualBox**. Multiple virtual machines were run simultaneously, and the CPU and memory utilization of the host system were monitored using Task Manager.